

25.1 Astronomy

Question Paper

Course	CIE A Level Physics
Section	25. Astronomy & Cosmology
Topic	25.1 Astronomy
Difficulty	Medium

Time allowed: 40

Score: /29

Percentage: /100

Question 1a

A star has a luminosity that is known to be $4.8 \times 10^{29} \text{ W}$. A scientist observing this star finds that the radiant flux intensity of light received on Earth from the star is 2.6 nW m^{-2} .

(i)

Name the term used to describe an astronomical object that has known luminosity.

[1]

(ii)

Determine the distance of the star from Earth.

distance = m [2]

[3 marks]**Question 1b**

The Sun has a surface temperature of 5800 K. The wavelength λ_{max} of light for which the maximum rate of emission occurs from the Sun is 500 nm.

The scientist observing the star in **(a)** finds that the wavelength for which the maximum rate of emission occurs from the star is 430 nm.

(i)

Show that the surface temperature of the star in **(a)** is approximately 6700 K. Explain your reasoning.

[2]

(ii)

Use the information in **(a)** and **(b)(i)** to determine the radius of the star.

radius = m [2]

[4 marks]

Question 2a

Table 1.1 summarises some information about four stars in the constellation Cygnus.

Table 1.1

Name	Colour	Luminosity / L_{Sun}
Aljanah	orange	62
Deneb	blue-white	196 000
Fawaris	blue	155
Sadr	yellow-white	33 000

Identify the star in Table 1.1 with

- (i)
the highest surface temperature, [1]
- (ii)
the lowest surface temperature, [1]
- (iii)
the largest radius. [1]

[3 marks]

Question 2b

Fig. 1.2 shows the rate of emission from two of the stars, Sadr and Aljanah, plotted against wavelength.

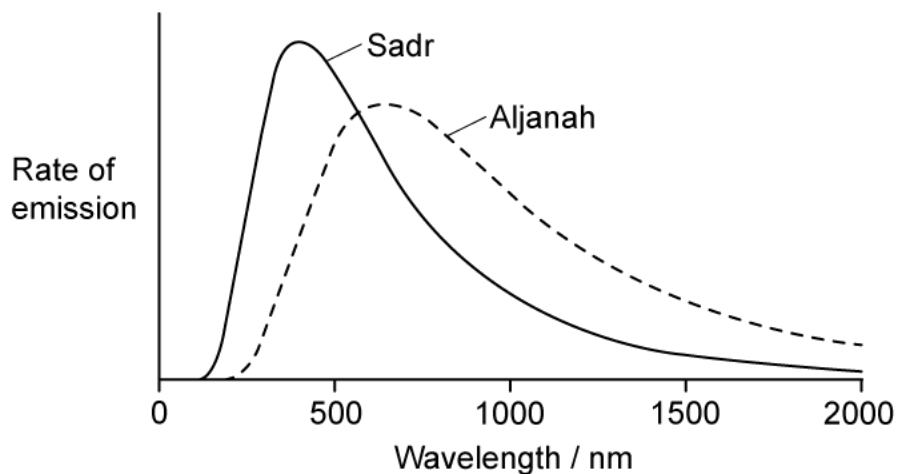


Fig. 1.2

Explain how the curves in Fig. 1.2 can be used to determine the surface temperature of the stars.

[3 marks]

Question 2c

The surface temperature of Aljanah is known to be 4700 K.

Determine the surface temperature of Sadr.

[3 marks]

Question 2d

Calculate the radius of Sadr in units of solar radius R_{Sun} .

Radius of the sun, $R_{\text{Sun}} = 6.96 \times 10^8 \text{ m}$

Luminosity of the Sun, $L_{\text{Sun}} = 3.83 \times 10^{26} \text{ W}$

[2 marks]

Question 3a

State what is meant by the luminosity of a star.

[1 mark]

Question 3b

In a comparison of two stars, A and B, the following data was collected

Surface temperature of star A = 25 000 K

Surface temperature of star B = 4300 K

The radius of star B was determined to be 1.1×10^5 times larger than the radius of star A.

Calculate the ratio

$$\frac{\text{*luminosity of star B*}}{\text{*luminosity of star A*}}$$

[3 marks]

Question 3c

The wavelength λ_{max} of light for which the maximum rate of emission occurs from the Sun is $5.00 \times 10^{-7} \text{ m}$.

The surface temperature of the Sun is 5770 K.

Determine the wavelengths of light for which the maximum rate of emission occurs from stars A and B.

[3 marks]

Question 3d

Fig. 1.1 shows the variation of rate of emission against wavelength for the Sun.

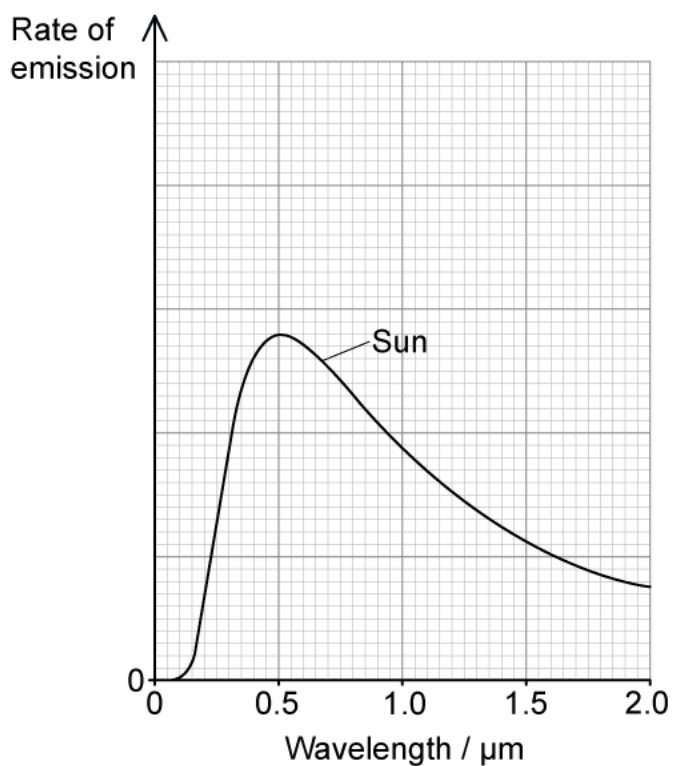


Fig. 1.1

On Fig. 1.1, sketch the variation of rate of emission against wavelength for stars A and B.

[4 marks]



Model Answers: Medium

1a

(a)

(i) An astronomical object with a known luminosity is:

- A standard candle; [1 mark]

(ii) Determine the distance of the star from Earth:

List the known quantities:

- Luminosity, $L = 4.8 \times 10^{29} \text{ W}$
- Radiant flux intensity, $F = 2.6 \text{ nW m}^{-2} = 2.6 \times 10^{-9} \text{ W m}^{-2}$

Use the inverse square law of flux relationship to determine the distance of the star from Earth, d :

- $F = \frac{L}{4\pi d^2}$ [1 mark]
- $d^2 = \frac{L}{4\pi F} \Rightarrow d = \sqrt{\frac{L}{4\pi F}}$
- $d = \sqrt{\frac{(4.8 \times 10^{29})}{4\pi \times (2.6 \times 10^{-9})}}$
- $d = 3.83 \times 10^{18} \text{ m} = 3.8 \times 10^{18} \text{ m (2 s.f.)}$ [1 mark]

[Total: 3 marks]

- It's easy to miss the **inverse square law of flux** relationship in the revision notes but make sure you learn it along with:
 - Wien's Displacement Law
 - Stefan-Boltzmann Law

1b

(b)

(i) Show that the surface temperature of the star in **(a)** is approximately 6700 K.
Explain your reasoning:

List the known quantities:

- Peak wavelength emitted by the Sun, $\lambda_{Sun} = 500 \text{ nm} = 500 \times 10^{-9} \text{ m}$
- Surface temperature of Sun, $T_{Sun} = 5800 \text{ K}$
- Peak wavelength emitted by the star, $\lambda_{max} = 430 \text{ nm} = 430 \times 10^{-9} \text{ m}$

Use Wien's displacement law to calculate the temperature T_{star} :

- $\lambda_{max} \propto \frac{1}{T}$ [1 mark]
- $\frac{\lambda_{Sun}}{\lambda_{star}} = \frac{T_{star}}{T_{Sun}}$
- $T_{star} = 5800 \times \frac{500}{430}$
- $T_{star} = 6740 \text{ K} = 6700 \text{ K} (2 \text{ s.f.})$ [1 mark]

(ii) Use the information in **(a)** and **(b)(i)** to determine the radius of the star:

List the known quantities:

- Luminosity, $L = 4.8 \times 10^{29} \text{ W}$
- Temperature, $T = 6700 \text{ K}$
- Stefan-Boltzmann Law: $L = 4\pi\sigma r^2 T^4$
- Stefan-Boltzmann Constant, $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Use the Stefan-Boltzmann Law:

- $L = 4\pi\sigma r^2 T^4$
- $r^2 = \frac{L}{4\pi\sigma T^4}$
- $r = \sqrt{\frac{(4.8 \times 10^{29})}{4\pi \times (5.67 \times 10^{-8}) \times 6700^4}}$ [1 mark]
- $r = 1.83 \times 10^{10} \text{ m} = 1.8 \times 10^{10} \text{ m (2s.f.)}$ [1 mark]

[Total: 4 marks]

- You need to use your **GCSE maths** skills to remember how to find the **constant of proportionality** for two quantities that are **directly proportional**

2a

(a) The star with the...

(i) highest surface temperature is:

- Fawaris; [1 mark]

(ii) lowest surface temperature is:

- Aljanah; [1 mark]

(iii) largest radius is:

- Deneb; [1 mark]

*Accept Sadr***[Total: 3 marks]**

Remember that **blue** is the **hottest** colour possible, while **red** is the **coolest**.

White comes **between** yellow and blue, so yellow-white is hotter than yellow, but blue-white is cooler than blue!

The radius of a star can be estimated using the Stefan-Boltzmann Law

$$L = 4\pi r^2 \sigma T^4 \Rightarrow r^2 \propto \frac{L}{T^4}$$

Since Deneb has the largest luminosity by far, you can expect it to have the largest radius, which it does ($\approx 200 R_{\text{Sun}}$).

However, due to the power of 4 on T^4 , the surface temperature of the star also has a significant impact on a star's radius, so Sadr would also have been an acceptable estimate.

Sadr does not have the largest radius, but it is close to the radius of Deneb since it has a much lower temperature ($\approx 150 R_{\text{Sun}}$).

2b

(b) The curves can be used to determine the surface temperature of the stars because:

- Wien's law states that the wavelength of maximum intensity (emitted by a star) is inversely proportional to its (thermodynamic) temperature **OR** $\lambda_{\text{max}} \propto \frac{1}{T}$; [1 mark]
- (To determine the surface temperature), read off the peak wavelength λ_{max} from the graph; [1 mark]
- Use a reference star or star of known λ_{max} and T to find a constant of proportionality; [1 mark]

[Total: 3 marks]

2c

(c) Determine the surface temperature of Sadr:

List the known quantities:

- Surface temperature of Aljanah, $T_A = 4700 \text{ K}$

Use the graph to determine the peak wavelengths:

[Total: 3 marks]

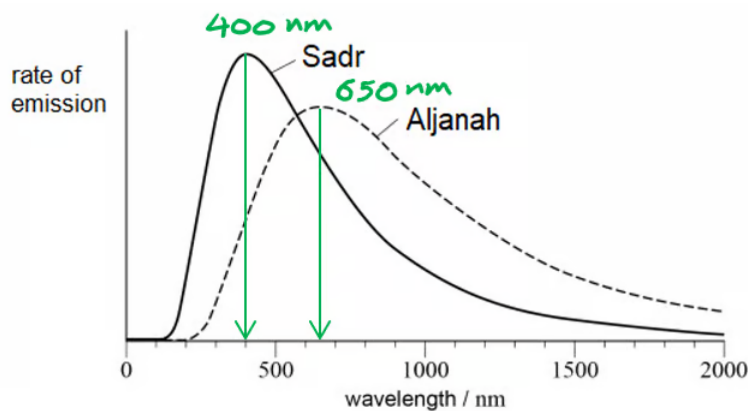
2c

(c) Determine the surface temperature of Sadr:

List the known quantities:

- Surface temperature of Aljanah, $T_A = 4700 \text{ K}$

Use the graph to determine the peak wavelengths:



- Aljanah: $\lambda_A = 650 \text{ nm}$ **AND** Sadr: $\lambda_S = 400 \text{ nm}$ [1 mark]

Use Wien's law to determine the surface temperature of Sadr:

- Wien's law: $T \propto \frac{1}{\lambda_{max}}$

- Therefore, $\frac{T_S}{T_A} = \frac{\lambda_A}{\lambda_S}$

- $T_S = T_A \times \frac{\lambda_A}{\lambda_S} = 4700 \times \frac{650}{400}$ [1 mark]

- $T_S = 7600 \text{ K}$ [1 mark]

[Total: 3 marks]

You must be confident in turning proportionalities into equations.

Since $T \propto \frac{1}{\lambda_{max}}$, this means $T\lambda_{max} = \text{constant}$. This constant is going to be the same for both stars.

$$\text{Therefore, } T_S \lambda_S = T_A \lambda_A \Rightarrow \frac{T_S}{T_A} = \frac{\lambda_A}{\lambda_S}.$$

2d

(d) Calculate the radius of Sadr in units of solar radius:

List the known quantities:

- Luminosity of Sadr, $L = 33\,000 L_{\text{Sun}} = 33\,000 \times (3.83 \times 10^{26}) \text{ W}$
- Surface temperature of Sadr, $T = 7600 \text{ K}$ (from **(c)**)
- Radius of the sun, $R_{\text{Sun}} = 6.96 \times 10^8 \text{ m}$
- Stefan-Boltzmann constant, $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

Use the Stefan-Boltzmann law to calculate radius:

$$\bullet \quad L = 4\pi r^2 \sigma T^4 \Rightarrow r = \sqrt{\frac{L}{4\pi \sigma T^4}}$$

$$\bullet \quad r = \sqrt{\frac{33\,000 \times (3.83 \times 10^{26})}{4\pi \times (5.67 \times 10^{-8}) \times (7600)^4}}$$

$$\bullet \quad \text{Radius: } r = 7.29 \times 10^{10} \text{ m [1 mark]}$$

Convert to units of R_{Sun} :

$$\bullet \quad \text{Radius: } r = \frac{7.29 \times 10^{10}}{6.96 \times 10^8} = 105 R_{\text{Sun}} \text{ [1 mark]}$$

[Total: 2 marks]

The luminosity of Sadr is given in Table 1.1. This is a good example of making sure you look at **all** the previous subparts of a question when attempting it, chances are you will be expected to use values from previous subparts.

3a

(a) Luminosity of a star is defined as:

- The total power of radiation emitted (by the star); [1 mark]

[Total: 1 mark]

3b

(b) Calculate the ratio of the luminosities of stars A and B:

List the known quantities:

- Surface temperature of star A, $T_A = 25\,000\text{ K}$
- Surface temperature of star B, $T_B = 4300\text{ K}$
- Ratio of radii $= \frac{r_B}{r_A} = 1.1 \times 10^5$

Use the Stefan-Boltzmann law to determine the ratio of the luminosities:

- Stefan-Boltzmann law: $L = 4\pi r^2 \sigma T^4$
- Ratio: $\frac{L_B}{L_A} = \frac{r_B^2 T_B^4}{r_A^2 T_A^4} = \left(\frac{r_B}{r_A}\right)^2 \times \frac{T_B^4}{T_A^4}$ [1 mark]
- $\frac{L_B}{L_A} = \frac{(1.1 \times 10^5)^2 \times (4300)^4}{(25\,000)^4}$ [1 mark]
- $\frac{L_B}{L_A} = 1.059 \times 10^7 = 1.1 \times 10^7$ [1 mark]

[Total: 3 marks]

In the equation $L = 4\pi r^2 \sigma T^4$, $4\pi\sigma$ is a **constant**. Therefore, $\frac{L}{r^2 T^4} = 4\pi\sigma = \text{constant}$

This means $\frac{L_B}{r_B^2 T_B^4} = \frac{L_A}{r_A^2 T_A^4}$ and $\frac{L_B}{L_A} = \frac{r_B^2 T_B^4}{r_A^2 T_A^4}$.

3c

(c) Determine the wavelengths of light for which the maximum rate of emission occurs from stars A and B:

List the known quantities:

- Surface temperature of the Sun, $T_{Sun} = 5770 \text{ K}$
- Max wavelength from the Sun, $\lambda_{Sun} = 5.00 \times 10^{-7} \text{ m}$
- Surface temperature of star A, $T_A = 25\,000 \text{ K}$
- Surface temperature of star B, $T_B = 4300 \text{ K}$

Use Wiens law to determine the constant of proportionality:

- Wien's law: $\lambda_{max} \propto \frac{1}{T}$ [1 mark]
- $T_{Sun} \lambda_{Sun} = \text{constant}$
- $T_{Sun} \lambda_{Sun} = 5770 \times (5.00 \times 10^{-7}) = 2.885 \times 10^{-3} \text{ (m K)}$

Determine the maximum wavelength from star A:

- Star A: $\lambda_A = \frac{2.885 \times 10^{-3}}{25\,000}$
- Star A: $\lambda_A = 1.154 \times 10^{-7} \text{ m}$ [1 mark]

Determine the maximum wavelength from star B:

- Star B: $\lambda_B = \frac{2.885 \times 10^{-3}}{4300}$

This means $\frac{L_B}{r_B^2 T_B^4} = \frac{L_A}{r_A^2 T_A^4}$ and $\frac{L_B}{L_A} = \frac{r_B^2 T_B^4}{r_A^2 T_A^4}$.

3c

(c) Determine the wavelengths of light for which the maximum rate of emission occurs from stars A and B:

List the known quantities:

- Surface temperature of the Sun, $T_{Sun} = 5770 \text{ K}$
- Max wavelength from the Sun, $\lambda_{Sun} = 5.00 \times 10^{-7} \text{ m}$
- Surface temperature of star A, $T_A = 25\,000 \text{ K}$
- Surface temperature of star B, $T_B = 4300 \text{ K}$

Use Wiens law to determine the constant of proportionality:

- Wien's law: $\lambda_{max} \propto \frac{1}{T}$ [1 mark]
- $T_{Sun} \lambda_{Sun} = \text{constant}$
- $T_{Sun} \lambda_{Sun} = 5770 \times (5.00 \times 10^{-7}) = 2.885 \times 10^{-3} \text{ (m K)}$

Determine the maximum wavelength from star A:

- Star A: $\lambda_A = \frac{2.885 \times 10^{-3}}{25\,000}$
- Star A: $\lambda_A = 1.154 \times 10^{-7} \text{ m}$ [1 mark]

Determine the maximum wavelength from star B:

- Star B: $\lambda_B = \frac{2.885 \times 10^{-3}}{4300}$
- Star B: $\lambda_B = 6.709 \times 10^{-7} \text{ m}$ [1 mark]

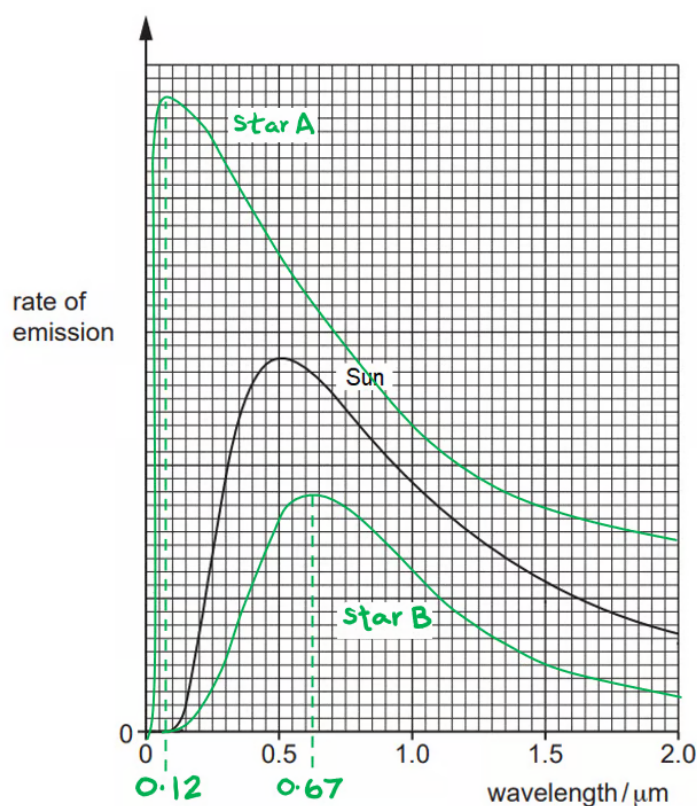
[Total: 3 marks]

3d

(d) The variation of rate of emission against wavelength for stars A and B is as follows:

List the known quantities:

- Maximum wavelength of star A: $\lambda_A = 1.154 \times 10^{-7} \text{ m} = 0.1154 \mu\text{m}$ (from (c))
- Maximum wavelength of star B: $\lambda_B = 6.709 \times 10^{-7} \text{ m} = 0.67 \mu\text{m}$ (from (c))



- Star A peak at $0.12 \mu\text{m}$; [1 mark]
- Star A curve shifted to the left (of the Sun); [1 mark]
- Star B peak at $0.67 \mu\text{m}$; [1 mark]
- Star B curve shifted to the right (of the Sun); [1 mark]

[Total: 4 marks]